FACTORING

BASIC ALGEBRA

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GRADING SYSTEM

FINAL GRADE

30%

PRELIM

30%

MIDTERM

40%

ENDTERM

TERM GRADE

70%

WEEKLY

30%

EXAM

WEEKLY GRADE

70%

LONG QUIZ

30%

SEATWORK

PENALTY

0

ABSENT

3

45-min LATE

7 max

15-min LATE

TOPIC OUTLINE

The Greatest Common Factor

Factoring by Grouping

Factoring Quadratics

Synthetic Division

THE GREATEST COMMON FACTOR

GREATEST COMMON FACTOR

Factor $\mathbf{15 + 25x}$

The **largest** expression that divides each term

Factor $x^2y + y^2x^3$

FACTORING BY GROUPING

FACTORING BY GROUPING

Factor $\boldsymbol{x^3 + 3x^2 + 4x + 12}$

Grouping terms (**4 terms**) with common factor

Factor $\mathbf{2x^3 - 10x^2 + 7x - 35}$

FACTORING QUADRATICS

Factor $\boldsymbol{x^2 - 6x + 8}$

A quadratic expression is a **second**-degree polynomial expression

Factor $\mathbf{10x^2 + 11x - 6}$

DIFFERENCE OF SQUARES

Factor $\boldsymbol{x^2 - 16}$

$$\boldsymbol{a^2 - b^2 = (a + b)(a - b)}$$

Factor $\mathbf{9p^2 - 1}$

Sum of squares **DOES NOT** factor

$$x^2 + 4 \neq x^2 + 2^2$$

DIFFERENCE OR SUM OF CUBES

Factor $y^3 + 27$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2)$$

Which of these **DOES NOT** factor?

A. $x^2 + x$

B. $x^2 - 25$

C. $x^2 + 4$

D. $x^3 + 2x^2 + 3x + 6$

E. $5x^2 - 14x + 8$

SYNTHETIC DIVISION

Factor $\boldsymbol{x^3 + 4x^2 - 11x - 30}$

A **shorthand method** of dividing polynomials for the special case of dividing by a **linear factor** whose leading coefficient is 1

SEATWORK